

BOLOGOYE, Russian Federation

WMO: 262980

Lat: 57.883N

Lon: 34.033E

Elev: 187

StdP: 99.10

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.4	-21.6	-27.7	0.3	-25.3	-24.0	0.4	-21.1	7.6	-2.4	6.4	-2.7	0.9	320	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	28.4	19.7	26.4	18.8	24.6	17.8	20.8	26.6	19.7	24.7	18.7	23.1	2.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.2	17.8	13.1	21.8	17.0	12.4	21.1	60.8	26.8	57.0	24.9	53.9	23.2	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.4	4.8	-29.3	30.5	3.3	2.2	-31.7	32.1	-33.6	33.4	-35.5	34.6	-37.9	36.2		
(5)			-29.4	22.2	3.3	1.5	-31.8	23.2	-33.7	24.1	-35.6	24.9	-38.0	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-7.6	-7.4	-2.2	5.2	11.4	15.5	18.4	16.0	10.7	4.7	-1.1	-5.0
	DBStd	10.35	6.97	6.95	4.73	4.81	4.80	3.81	3.35	3.52	3.73	4.16	5.12	6.45
	HDD10.0	2615	545	487	379	159	40	2	0	1	36	171	332	464
	HDD18.3	4961	803	720	637	395	219	102	39	89	230	423	582	722
	CDD10.0	779	0	0	0	14	84	168	262	188	57	6	0	0
	CDD18.3	83	0	0	0	0	5	18	42	18	1	0	0	0
	CDH23.3	684	0	0	0	1	47	153	329	147	7	0	0	0
	CDH26.7	158	0	0	0	0	8	32	83	34	0	0	0	0
Wind	WSAvg	2.1	2.3	2.3	2.4	2.1	2.1	1.8	1.6	1.7	1.9	2.2	2.4	2.5
Precipitation	PrecAvg	681	49	36	35	35	58	78	87	79	59	63	53	49
	PrecMax	876	89	57	59	65	103	148	276	157	132	127	107	78
	PrecMin	423	14	11	8	13	23	21	18	6	18	15	3	17
	PrecStd	105	17	13	12	14	24	33	58	38	23	29	23	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	4.5	12.1	22.8	27.7	29.9	32.1	30.2	24.5	17.4	9.2	7.1
		MCWB	3.4	2.8	6.2	13.5	18.4	19.5	21.7	21.2	17.7	13.2	8.0	6.6
	2%	DB	2.1	2.9	8.1	19.1	24.3	27.3	28.9	26.8	20.6	13.5	7.7	4.9
		MCWB	1.3	1.6	3.6	11.1	16.2	18.7	20.2	18.9	15.4	11.1	6.8	4.4
	5%	DB	1.2	1.7	5.7	16.2	21.8	24.9	26.7	24.6	18.2	11.9	6.3	3.3
		MCWB	0.5	0.7	2.7	9.6	14.6	17.6	19.5	18.1	14.2	10.4	5.5	2.6
	10%	DB	0.5	1.0	3.8	13.6	19.6	22.8	24.8	22.5	16.4	10.5	5.0	2.0
		MCWB	-0.1	0.2	1.8	8.2	13.4	16.8	18.4	17.2	13.4	9.3	4.3	1.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.4	6.7	14.0	19.7	21.2	22.5	22.0	18.1	13.4	8.5	6.4
		MCDB	4.0	4.1	11.8	22.1	25.8	26.7	29.2	28.7	23.0	15.8	9.0	7.0
	2%	WB	1.6	1.7	4.5	11.8	17.6	19.6	21.1	20.2	16.2	11.7	6.9	4.5
		MCDB	1.9	2.5	6.8	17.2	22.7	25.2	27.4	25.0	19.3	13.1	7.5	4.9
	5%	WB	0.7	0.9	3.1	10.3	15.8	18.7	20.1	18.9	15.0	10.6	5.6	2.7
		MCDB	1.0	1.6	5.1	15.2	20.2	23.7	25.4	22.9	17.4	11.8	6.2	3.2
	10%	WB	0.0	0.2	2.0	8.8	14.2	17.4	19.1	17.7	13.7	9.3	4.3	1.5
		MCDB	0.5	0.9	3.6	13.2	18.4	21.5	23.4	21.5	16.1	10.5	4.9	1.9
Mean Daily Temperature Range	5% DB	MDBR	5.0	6.4	7.5	9.1	9.8	9.1	9.2	8.8	7.8	5.3	3.8	4.0
		MCDBR	3.4	4.3	10.2	13.3	12.9	12.1	11.9	12.2	10.8	7.4	4.6	3.3
		MCWBR	3.3	3.7	6.8	7.0	5.9	5.3	4.7	5.4	6.0	5.0	4.3	3.4
	5% WB	MCDBR	3.5	3.7	8.2	12.3	11.5	10.2	10.8	10.4	9.3	6.3	4.3	3.3
		MCWBR	3.5	3.4	5.7	6.6	5.7	4.9	4.8	5.0	5.5	4.6	4.1	3.3
Clear-Sky Solar Irradiance	taub	0.276	0.313	0.332	0.379	0.372	0.363	0.393	0.392	0.356	0.336	0.308	0.265	
	taud	2.123	2.106	2.169	2.256	2.386	2.435	2.362	2.353	2.433	2.403	2.235	2.111	
	Ebn at Noon	554	699	805	827	862	876	836	807	785	690	518	446	
	Edh at Noon	53	85	105	115	108	104	110	104	83	65	48	39	
All-Sky Solar Radiation	RadAvg	0.41	1.17	2.63	3.94	5.12	5.28	5.16	4.08	2.54	1.09	0.37	0.22	
	RadStd	0.08	0.21	0.33	0.45	0.45	0.50	0.66	0.37	0.34	0.22	0.06	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page